StegoVault: Deep Learning Steganography Platform

Technical Manual & Architectural Overview

1. Executive Summary

StegoVault is a full-stack steganographic platform designed to facilitate secure, deniable, and high-capacity communication by hiding sensitive payloads inside everyday carrier media. By utilizing deep learning architectures (including Convolutional Neural Networks and Transformers), the platform achieves imperceptible noise embedding that easily bypasses standard stego-detectors. The platform is built around a secure FastAPI backend, a high-fidelity React frontend, and is containerized with Docker to enable seamless GPU-accelerated computing and CPU fallback orchestration.

2. Neural Model Architectures and Modalities

StegoVault operates over four primary carrier media modalities: Image, Audio, Video, and Text. Each modality routes processing through a dedicated PyTorch model configured in the backend runtime. The model properties and validation bounds are configured as follows:

Modality	PyTorch Model Class	Max Limit	Simulation Bypass
Image	ImageStegoNet (2D CNN)	10 MB	99.4% Safe
Audio	AudioStegoNet (1D CNN)	20 MB	98.7% Safe
Video	VideoStegoNet (3D CNN)	50 MB	97.2% Safe
Text	TextStegoTransformer	2 MB	99.1% Safe

3. AI Engine Health & Auto-Scaling Hardware

To satisfy the computational demands of multi-modal neural networks, StegoVault includes a real-time AI Engine Health monitor. The system automatically queries hardware configurations on boot. When an NVIDIA GPU is available, the PyTorch models load into CUDA, allocating tensor operations dynamically. If a GPU is missing, the backend reports a CPU fallback, generating custom notifications. The dashboard telemetry monitors performance constants such as processing latency (ms), overall stego throughput (MB/s or GB/s), and active node uptime, ensuring high reliability.

4. Cryptographic Security and Signature Bound Key Model

Beyond structural media transformations, StegoVault implements a robust multi-layered security protocol. When an asset is encoded, the payload data is first encrypted using AES-GCM. A key-binding signature is generated using SHA-256(access_key + file_signature), where the file signature is a SHA-256 hash derived from the post-processed carrier media structure. This binding is packaged directly in the stego container payload. During decryption, the receiver must provide both the stego carrier file and the access key. If the key binding validation fails or the stego file has been altered by even one bit, the system rejects the decryption job immediately, preventing tampering attacks and visual detection.

5. Secure Vault Manager & Shard Analytics

The Secure Vault Manager is a local repository that manages all completed steganographic carrier media and extracted payloads. The vault is powered by SQLite query binding. All file entries display actual file sizes queried dynamically from the filesystem, along with interactive category filters. Selecting modality cards (Image, Audio, Video, or Document) filters the sealed asset registry. Clicking on individual records triggers the Details Telemetry Overlay. This modal displays critical stego transaction logs, operation types, timestamps, the active processor, a copyable AES access key, and the original embedded payload data for verification.

6. Audit Logging and Local History Database

To maintain operational secrecy, the platform includes a local history logging system. Every encoding, decoding, or validation event is stored in the local audit log. Clicking 'View History' opens the Operations History Modal. This lists all job IDs, modality types, source carrier file names, status indicators, and formatted 12-hour timestamps (e.g. 12:14:02 PM). Users can clear the entire audit trail using the confirmation-bound Clear History button, which purges all activity logs from local storage, removing trace logs of steganography operations.

7. Technical Conclusion

StegoVault integrates modern deep learning models with state-of-the-art cryptographic validation. The implementation of dynamic telemetry monitors, interactive modality hubs, and file-bound AES-GCM key signatures provides a highly secure, deployable, and transparent solution for covert data transmission, serving as an exceptional foundation for advanced security applications.